

ECON 712: Applied Econometrics

Lecture 6: Linear Regression using Vector Spaces, part II

Vector Space Definitions

- We use set of vectors to formally define the following subspaces:

1. k -dimensional subspace:

$$\mathcal{S}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k) \equiv \left\{ \mathbf{z} \in E^n \mid \mathbf{z} = \sum_{i=1}^k b_i \mathbf{x}_i, b_i \in \mathbb{R} \right\}$$

2. $(n - k)$ -dimensional orthogonal complement:

$$\mathcal{S}^\perp(\mathbf{w}) \equiv \{ \mathbf{w} \in E^n \mid \mathbf{w}^T \mathbf{z} = 0 \text{ for all } \mathbf{z} \in \mathcal{S}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k) \}$$

- We will also make use of the following inequality, called Cauchy-Schwartz:

$$\langle \mathbf{x}, \mathbf{y} \rangle \leq \|\mathbf{x}\| \|\mathbf{y}\|$$

Linear Regression

- Linear Regression considers each linearly-independent k column vector of size n from $\mathbf{X}$ as the basis vector that spans the k -dimensional space

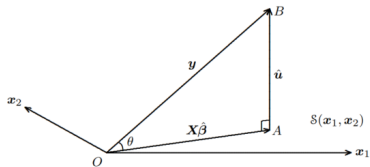
- ▶ We also assume the following two conditions:

$$\mathbf{X}^T \hat{\mathbf{u}} = \mathbf{0} \quad , \quad \mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}}$$

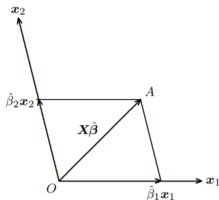
- ▶ This result in linear regression being defined as selecting $\hat{\boldsymbol{\beta}}$ that reaches the perpendicular dropped from $\mathbf{y}$ vector into the plane spanned by $\mathbf{X}$

Linear Regression

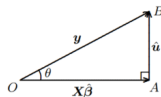
Example with $n = 3$, $k = 2$



(a) y projected on two regressors



(b) The span $S(x_1, x_2)$ of the regressors



(c) The vertical plane through y

Linear regression in three dimensions

Linear Regression

- The concept of dropping a perpendicular is formalized by an orthogonal projection:

$$\mathbf{P}_X = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \quad , \quad \mathbf{X}\hat{\beta} = \mathbf{P}_X \mathbf{y}$$

- ▶ An orthogonal projection is a matrix that projects any vector to a point in the subspace $\mathcal{S}(\mathbf{X})$ that it is closest to it
 - ▶ $\mathcal{S}(\mathbf{X})$ is called the invariant subspace of $\mathbf{P}_X$ because the projection of any vector in $\mathcal{S}(\mathbf{X})$ using $\mathbf{P}_X$ leaves that vector unchanged
- We are also going to use a projection onto the orthogonal subspace of $\mathcal{S}(\mathbf{X})$, $\mathcal{S}^\perp$

$$\mathbf{M}_X = \mathbf{I} - \mathbf{P}_X \quad , \quad \hat{\mathbf{u}} = \mathbf{M}_X \mathbf{y}$$

Linear Regression

- Important properties of the projection matrices:

1. Idempotent:

$$\mathbf{P}_X \mathbf{P}_X = \mathbf{P}_X$$

$$\mathbf{M}_X \mathbf{M}_X = \mathbf{M}_X$$

2. Annihilation: $\mathbf{P}_X \mathbf{M}_X = \mathbf{M}_X \mathbf{P}_X = \mathbf{0}$

- Also, by Pythagorean Theorem (Sum Squares):

$$\|\mathbf{y}\|^2 = \|\mathbf{P}_X \mathbf{y}\|^2 + \|\mathbf{M}_X \mathbf{y}\|^2$$

Linear Transformations

- Let $\mathbf{A}$ be a $k \times k$ matrix with full rank, and define $\mathbf{Z} = \mathbf{X} \mathbf{A}$ then :

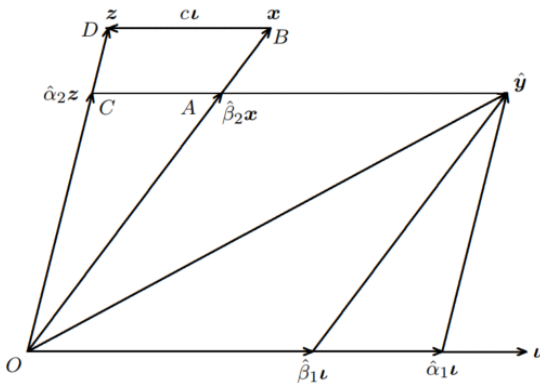
$$\mathbf{A} \hat{\alpha} = \hat{\beta}$$

where $\mathbf{X} \hat{\beta} = \mathbf{P}_X \mathbf{y}$, $\mathbf{Z} \hat{\alpha} = \mathbf{P}_Z \mathbf{y}$

- There are two cases (can be combined as in the Celsius/Fahrenheit example in ETM):
 - When the regressors are multiplied by a constant (e.g. change of units from cm to meters)
 - When a constant is added/subtracted from regressors

Linear Transformations

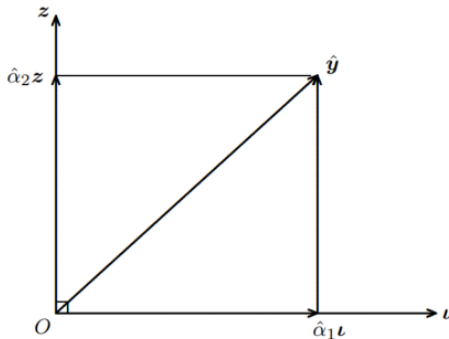
- Subtracting a constant ι_C from regressors when the constant term is included



Adding a constant does not affect the slope coefficient

Linear Transformations

- Subtracting a constant $\iota\bar{x}$ from regressors when the constant term is included



Orthogonal regressors may be omitted